Layer-Wise Attribute Sensitivity and Adaptive Fusion with Mixture of Experts in Deep Models for Color and Texture Classification

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***Abstract*—This study addresses the challenge of effectively extracting image content based on specific visual attributes, i.e., color, texture, or their combination; and proposes an adaptive fusion strategy based on Mixture of Experts (MoE). Experiments are conducted with five architectures (ResNet-50, VGG-16, iBOT, VMamba, and LMFCN) on a fashion dataset comprising 1,000 color images and 1,000 texture images. *Experiment 1* confirms that early layers are more effective for color classification, while deeper layers excel at texture recognition, a pattern that generalizes from classical CNNs to Vision Transformers and State Space Models. *Experiment 2* shows that the concatenation of early and deep layer features generally does not outperform specialized single-layer representations. *Experiment 3* introduces a trainable MoE model that, via a learned router, adaptively weights the contributions of the early (color-sensitive) and deep (texture-sensitive) layers on an image-basis. The MoE outperforms both Experiment 1 and Experiment 2 in nearly all architectures, achieving up to 98.13% with VMamba on texture data and 98.13% with iBOT on color data — improvements of 3.80 and 2.27 percentage points over the best individual layers, respectively.**

***Index Terms*—Convolutional Neural Networks, Deep Models, Visual Attributes, Vision Transformers, State Space Models, Mixture of Experts, Image Retrieval.**

I. Introduction

Content-Based Image Retrieval (CBIR) has broad applications in e-commerce, medical imaging, and surveillance [1], [9]. A key challenge is extracting relevant visual attributes, i.e., color, texture, or their combination, to match user queries (Figure 1).

Color and texture feature extraction has been a central topic in Content-Based Image Retrieval (CBIR). Classical color descriptors include color histograms and color moments, which capture statistical properties such as mean, variance, and skewness of color distributions [15]. For texture representation, methods such as Gabor filters and Local Binary Patterns (LBP) have been widely used [16].

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Fig. 1. Six different color and texture-based patterns (adapted from [2]).

With the advent of deep learning, Baloian et al. [2] demonstrated that different CNN layers exhibit attribute-specific sensitivity, with early layers being more responsive to color information and deeper layers capturing texture patterns. However, the behavior of attribute-specific representations across modern architectures (Vision Transformers and State Space Models) remains underexplored, and adaptive fusion strategies between specialized layers are still scarce in the literature.

In this work, we evaluate and compare classical and modern deep architectures for color- and texture-based image classification. We analyze layer-wise representations to investigate their sensitivity to visual attributes and the effectiveness of different fusion strategies. As a main contribution, we propose an adaptive fusion model based on Mixture of Experts (MoE) that learns, on an image-basis approach, the relative weights of the early (color) and deep (texture) layers through a trainable router.

The dataset [2] consists of fashion images divided into a **color subset** (1,000 images, 10 color classes) and a **texture subset** (1,000 images, 10 texture pattern classes).

The main contributions of this work are:

- A comprehensive layer-wise analysis of attribute sensitivity across classical CNNs, Vision Transformers, and State Space Models;
- A comparative benchmark of VGG-16, ResNet-50, iBOT, VMamba, and LMFCN for color- and texture-based image classification;

- Experimental demonstration that simple concatenation fusion does not improve performance for attribute-specific tasks;
- Proposal and evaluation of an MoE model that performs adaptive fusion of early and deep layer features, outperforming both concatenation and single-layer representations in most settings.

II. Related Work

The use of convolutional neural networks for representing object similarity has been extensively explored. Baloian et al. [2] investigated the behavior of ResNet-50 layers for color and texture-based retrieval, showing that early convolutional layers are more sensitive to color, while deeper layers better capture texture patterns. Features were extracted from ResNet-50 pre-trained on ImageNet [3], with Global Average Pooling (GAP) applied to each residual block and k-nearest neighbors (kNN) used for classification with 5 repetitions of stratified 70/30 holdout.

Zeiler and Fergus [20] showed that early CNN layers encode edges and colors while deep layers capture structural patterns; Kazami et al. [21] reported analogous behavior in ViTs. Unlike these qualitative studies, this work provides a systematic quantitative evaluation across CNNs, ViTs, and SSMs, and proposes a trainable adaptive fusion grounded in these findings.

Representation learning for attribute recognition has been explored via neural networks [7], [8] and Siamese architectures [6]. Classical CBIR descriptors include wavelet-based texture features [10] and color co-occurrence matrices [11]. Multi-layer fusion strategies — concatenation, late fusion, and attention-based combination [17], [18] — have been proposed, but their effectiveness for attribute-specific tasks remains limited.

Recent architectures include Mamba/VMamba [12] (SSM, linear time), iBOT [13] (self-supervised ViT), and LMFCN [19] (FCN with large-margin SVM). MoE [22] uses a router to weight local experts; Cao et al. [23] applied it to multimodal fusion. Its use for depth-wise layer selection within a single backbone remains unexplored — the gap this work addresses.

III. Evaluated Backbone Models

This section describes the five backbone architectures used in the experiments. Table I summarizes their key properties and the early/deep layer configuration used in the MoE model.

TABLE I
Summary of evaluated backbone architectures and MoE layer configuration (E = early, D = deep).

Backbone	**Type**	**Feat.**	**Dims**	**MoE E/D**
VGG-16	CNN	GAP/block	64–512	Bl.1/Bl.3
ResNet-50	CNN	GAP/res.blk	256–2048	RB1/RB3
iBOT	ViT	CLS token	384	Bl.2/Bl.9
VMamba	SSM	GAP/stage	96–768	St.1/St.2
LMFCN	FCN+SVM	SVM layer	640–2560	–

a) ResNet-50: ResNet-50 [4] is a 50-layer CNN with identity-based skip connections. Features are extracted from each residual block via GAP.

b) VGG-16: VGG-16 [5] stacks 3×3 convolutional layers across 5 blocks. Features are extracted via GAP at each block.

c) VMamba: VMamba [12] is a Structured State Space Model (SSM) that achieves linear-time processing via a selective scanning mechanism. We use the VMamba-Tiny variant with features extracted via GAP at each stage.

d) iBOT: iBOT [13] is a self-supervised ViT framework using masked image modeling with an online tokenizer (ViT-S/16, 12 blocks, CLS token).

e) LMFCN: LMFCN [19] trains FCNs optimized for large-margin SVM classification. We evaluate TCNN (3–6 layers) and ResNet-18 variants. Due to the tight SVM–FCN coupling, LMFCN is evaluated under Experiments 1 and 2 only.

IV. Experimental Methodology

A. Dataset and Protocol

The dataset was introduced in [2] and consists of 10 color classes and 10 texture classes with 100 images per class, totaling 1,000 images per task. Data were split into 70% training and 30% test in 5 repetitions of stratified holdout. Following the protocol established in [2] and widely adopted in the attribute-based CBIR literature, feature vectors are evaluated with a k-nearest neighbor (kNN) classifier ($k = 5$, Euclidean distance). For retrieval evaluation, the training set serves as the gallery to ensure disjoint query and database sets, avoiding trivial matches.

B. Layer-Wise Sensitivity Analysis

For each architecture, features are extracted from each individual layer or processing block. CNNs use GAP to produce a single feature vector per layer; iBOT uses the CLS token; VMamba applies GAP to each stage's feature map. For LMFCN, we evaluate TCNN variants (3–6 layers) and ResNet-18 with 2 or 4 residual blocks.

C. Early–Deep Concatenation Fusion

For each backbone, features from the best color layer (E) and the best texture layer (D) are concatenated: $z = [f_E \ ; \ f_D]$, and the resulting vector is evaluated with kNN.

D. Adaptive Fusion with Mixture of Experts

The proposed MoE model learns, per image, adaptive weights α and β for the early and deep layers, respectively.

Architecture. For each backbone, two projection heads h_E and h_D — each comprising a linear layer, LayerNorm, and ReLU — map heterogeneous-dimensional features to a common space of dimension $d = 256$. A Router (two-layer MLP with Softmax) receives the concatenation $[h_E(f_E); h_D(f_D)]$ and produces weights α and β with $\alpha + \beta = 1$. The final embedding is given by:

$$z = \alpha \cdot h_E(f_E) + \beta \cdot h_D(f_D). \quad (1)$$

The backbone is frozen during training; only h_E, h_D, and the Router are updated. An auxiliary linear head supervises training with cross-entropy loss; at evaluation time, only the embeddings z are used with kNN.

Training protocol. The training process considered AdamW optimizer, $lr = 10^{-4}$, weight decay $= 10^{-5}$, CosineAnnealingLR with $T_{\max} = 30$ epochs, batch size 32, basic data augmentation (random 224×224 crop and horizontal flip) during training only. Final evaluation: kNN with $k = 5$ and StandardScaler on embeddings z, under the same 5-repetition holdout protocol.

Table II summarizes the early and deep layer selections per backbone in the MoE model.

TABLE II
EARLY AND DEEP LAYER CONFIGURATION FOR EACH BACKBONE IN THE MOE MODEL.

Backbone	Early (E)	Dim E	Deep (D)	Dim D
VGG-16	Block 1	64	Block 3	256
ResNet-50	Res. Block 1	256	Res. Block 3	1024
iBOT	Block 2	384	Block 9	384
VMamba	Stage 1	192	Stage 2	384

V. EXPERIMENTAL RESULTS

Results cover layer-wise sensitivity (Sections V-A and V-B), fusion strategies (V-C and V-D), and CBIR evaluation using mAP@10, R@1, and R@5 (V-E).

A. Layer-Wise Sensitivity

Table III summarizes the best layer per backbone for color and texture classification. Across all architectures, early layers consistently achieve higher color accuracy, while deeper layers are more effective for texture — confirming and extending the pattern originally reported for ResNet-50 in [2]. In VGG-16, Layer 1 peaks at 95.13% for color, while Layer 4 reaches 94.73% for texture. ResNet-50 follows the same trend: Residual Block 2 for color (95.87%) and Block 3 for texture (93.27%). iBOT shows the steepest gradient across its 12 transformer blocks — color accuracy falls from 94.53% at Block 0 to 62.80% at Block 11, while texture climbs to 97.20% at Block 9, the highest value among all evaluated architectures. VMamba exhibits a more compressed version of the same pattern across its four stages, with Stage 1 best for color (94.80%) and Stage 2 for texture (94.33%).

Figure 2 illustrates the full depth-accuracy curves, showing the inverse pattern between color and texture per backbone.

B. Layer-Wise Sensitivity — LMFCN

Table IV presents LMFCN results. On the texture dataset, increasing TCNN depth consistently improves performance. On the color dataset, higher accuracy is obtained from earlier layers. LMFCN with its SVM classifier achieved 96.87%±0.9 on texture (ResNet-18 with 2 blocks) and 97.73%±0.95 on color (TCNN3).

TABLE III
BEST LAYER PER BACKBONE FOR COLOR AND TEXTURE CLASSIFICATION (%). BOLD: HIGHEST VALUE PER ATTRIBUTE COLUMN.

Backbone	Best Color	Color	Std	Best Tex.	Tex.	Std
VGG-16	Layer 1	95.13	0.81	Layer 4	94.73	0.74
ResNet-50	RB 2	**95.87**	0.50	RB 3	93.27	0.25
iBOT	Block 2	**95.87**	1.05	Block 9	**97.20**	0.54
VMamba	Stage 1	94.80	0.62	Stage 2	94.33	1.40

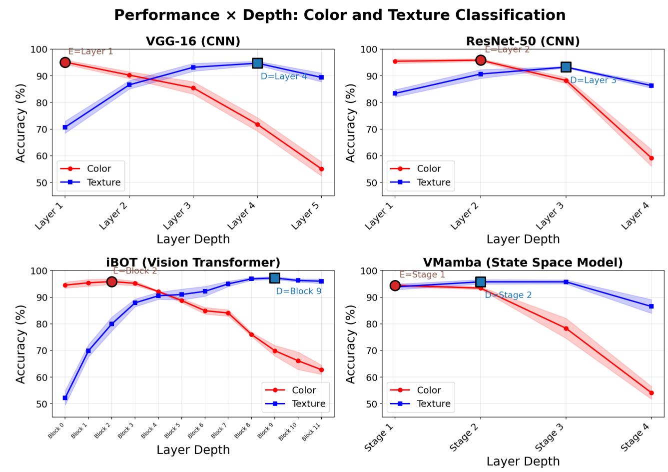

Fig. 2. Performance vs. depth: color and texture classification for VGG-16, ResNet-50, iBOT, and VMamba. Markers indicate the best layer per attribute (E=color, D=texture).

C. Results of Early–Deep Concatenation Fusion

Table V presents the results of the concatenation fusion for both color and texture tasks, where E denotes the best color layer, D the best texture layer, and Z the merged representation.

Fusion results show that simple concatenation generally does not improve classification performance. For color classification, fusion consistently degraded accuracy compared to using the specialized color layer alone (E), with iBOT showing the largest degradation (−21.53%). For texture classification, results were mixed, with minimal improvement for ResNet-50 (+0.27%) and minor degradation for iBOT (−0.20%) and VMamba (−0.93%).

TABLE IV
ACCURACY OF RESNET-18 CONVOLUTIONAL BLOCK (CB) AND RESIDUAL BLOCKS (RB) AND TCNN LAYERS USING LMFCN.

Method	Color	Std	Tex.	Std	Dim
TCNN3 Layer 3	**95.94**	0.43	**58.27**	1.30	640
TCNN4 Layer 4	**94.80**	0.65	**60.40**	2.60	640
TCNN5 Layer 5	95.40	0.96	**62.93**	1.65	640
TCNN6 Layer 6	94.53	0.84	**64.34**	2.69	640
ResNet-18(2) RB1	**96.13**	0.73	84.87	1.38	640
ResNet-18(2) RB2	91.93	0.98	**91.00**	1.62	1280
ResNet-18(4) RB1	**95.93**	0.98	84.13	1.59	640
ResNet-18(4) RB3	89.53	1.37	**92.47**	1.48	2560

TABLE V
CONCATENATION FUSION RESULTS (%). BOLD: BEST VALUE PER ROW.

Backbone	Task	E	D	Z	Δ
VGG-16	Color	**95.13**	71.80	86.13	-9.00
	Texture	70.73	**94.73**	94.73	$+0.00$
ResNet-50	Color	**95.87**	88.27	94.13	-1.73
	Texture	90.67	93.27	**93.53**	$+0.27$
iBOT	Color	**95.87**	69.93	74.33	-21.53
	Texture	79.93	**97.20**	97.00	-0.20
VMamba	Color	**94.80**	93.93	94.13	-0.67
	Texture	88.40	**94.33**	93.40	-0.93

D. Results of Adaptive MoE Fusion

Table VI presents the results of the proposed MoE model. Compared to single-layer extraction and concatenation fusion, the MoE outperforms both in nearly all backbone-attribute combinations.

TABLE VI
MOE MODEL RESULTS FOR COLOR AND TEXTURE CLASSIFICATION.

Backbone	MoE Color (%)	Std	MoE Tex. (%)	Std
VGG-16	94.20	1.68	**95.80**	1.63
ResNet-50	95.87	0.88	**97.73**	0.65
iBOT	**98.13**	0.69	94.80	1.17
VMamba	97.20	0.45	**98.13**	0.62

Table VII consolidates the comparison across all three strategies.

TABLE VII
COMPARISON ACROSS SINGLE-LAYER (EXP. 1), CONCATENATION (EXP. 2), AND MOE (EXP. 3). VALUES IN %. Δ_1: GAIN OVER SINGLE LAYER; Δ_2: GAIN OVER CONCATENATION.

Config.	Exp.1	Exp.2	MoE	Δ_1	Δ_2	Best
VGG Color	95.13	86.13	94.20	-0.93	$+8.07$	Exp.1
VGG Tex.	94.73	94.73	95.80	$+1.07$	$+1.07$	**MoE**
ResNet Col.	95.87	94.13	95.87	$+0.00$	$+1.73$	Exp.1
ResNet Tex.	93.27	93.53	97.73	$+4.47$	$+4.20$	**MoE**
iBOT Color	95.87	74.33	98.13	$+2.27$	$+23.80$	**MoE**
iBOT Tex.	97.20	97.00	94.80	-2.40	-2.20	Exp.1
VMamba Col.	94.80	94.13	97.20	$+2.40$	$+3.07$	**MoE**
VMamba Tex.	94.33	93.40	98.13	$+3.80$	$+4.73$	**MoE**

The MoE achieves the best or tied performance in 6 out of 8 evaluated configurations. The two exceptions, i.e., VGG-16 color and iBOT texture, correspond to cases where a single specialized layer already achieves very high performance (95.13% and 97.20%, respectively), leaving little room for improvement. The MoE gain over concatenation is expressive in 7 of 8 configurations, particularly for iBOT color (+23.80 pp) and VMamba texture (+4.73 pp); the exception is iBOT texture, where concatenation already preserves the discriminative capacity of Block 9 (97.00%).

E. CBIR Evaluation: mAP@10, R@1 and R@5

Table VIII reports the retrieval evaluation of each embedding under a standard CBIR protocol: for each test query, all training images are ranked by Euclidean distance and metrics are computed over the $K = 10$ nearest neighbors. mAP@10 is the mean average precision over the top-10 retrieved items; R@1 and R@5 indicate, respectively, the fraction of queries for which the correct class appears at rank 1 and within the top 5.

TABLE VIII
CBIR EVALUATION — COLOR AND TEXTURE (%). BEST VALUE PER BACKBONE AND TASK IN **BOLD**.

BB	Task	Method	mAP	R@1	R@5
VGG-16	Col.	Early (E)	**94.47**	**95.47**	**99.13**
		Deep (D)	72.92	72.20	91.53
		Concat.	83.89	86.53	96.73
		MoE	93.71	94.93	98.07
	Tex.	Early (E)	70.87	70.33	91.27
		Deep (D)	94.78	96.20	**99.47**
		Concat.	94.53	96.13	99.27
		MoE	**95.98**	**96.80**	99.07
ResNet-50	Col.	Early (E)	94.51	96.13	99.00
		Deep (D)	86.25	87.67	97.20
		Concat.	92.40	93.93	98.33
		MoE	**96.09**	**96.07**	**98.67**
	Tex.	Early (E)	90.35	90.80	99.07
		Deep (D)	93.72	95.60	99.20
		Concat.	93.05	94.47	98.87
		MoE	**97.88**	**98.13**	**99.20**
iBOT	Col.	Early (E)	94.93	95.40	99.13
		Deep (D)	71.85	72.00	92.47
		Concat.	75.06	76.67	94.13
		MoE	**97.96**	**98.27**	**99.47**
	Tex.	Early (E)	78.90	79.33	94.33
		Deep (D)	**96.40**	**97.73**	**99.80**
		Concat.	96.30	97.67	99.73
		MoE	95.10	95.40	98.53
VMamba	Col.	Early (E)	93.20	94.33	98.47
		Deep (D)	91.73	93.13	98.33
		Concat.	92.51	93.67	98.40
		MoE	**96.91**	**96.87**	**99.07**
	Tex.	Early (E)	87.66	88.80	97.33
		Deep (D)	93.52	95.53	99.13
		Concat.	93.05	95.07	99.13
		MoE	**97.89**	**98.27**	**99.33**

The MoE achieves the best CBIR metrics in 3 out of 4 backbones for color and 3 out of 4 for texture. The exception for iBOT texture mirrors the accuracy result: Block 9 produces texture representations so discriminative (mAP@10 = 96.40%) that the router cannot improve upon them. The MoE gain over concatenation is particularly expressive for iBOT color (+22.90 pp in mAP@10), confirming that adaptive fusion resolves the degradation observed in Experiment 2.

F. Qualitative Retrieval Analysis

Figures 3 and 4 show representative retrieval examples using the iBOT backbone. Each row shows a query image (leftmost, grey border) and its top-5 nearest neighbors ranked by Euclidean distance in the feature space; green borders indicate items of the correct class and red borders indicate errors.

For color retrieval (Block 2), the model consistently returns same-color garments regardless of clothing type — e.g., all five blue-query neighbors are blue items despite covering t-shirts,

Fig. 3. Top-5 retrieval examples for color classification (iBOT Block 2, fold 1). Queries: yellow, blue, grey, purple, black. Green = correct class; red = wrong class.

hoodies, and jackets. The near-perfect visual coherence of the top-5 results reflects the high mAP@10 (94.93%) obtained at this layer.

Fig. 4. Top-5 retrieval examples for texture classification (iBOT Block 9, fold 1). Queries: argyle, striped, floral, leopard, checkered. Green = correct class; red = wrong class.

For texture retrieval (Block 9), the deep transformer representations successfully cluster structural patterns: argyle diamonds, floral motifs, and leopard spots are retrieved with high visual fidelity. The few errors visible involve visually ambiguous patterns (e.g., dense stripes confused with checkered), which are the same failure modes observed in the mAP@10 metric (96.40% at Block 9).

VI. Discussion

The layer-wise sensitivity pattern — early layers for color, deep layers for texture — extends from classical CNNs [20] to ViTs [21] and, as shown here, to VMamba (SSM). iBOT's exceptional texture accuracy (97.20%) indicates that self-supervised masked modeling fosters rich structural representations; the uniform 384d dimensionality across iBOT blocks suggests representation quality, not size, is the determining factor.

Experiment 2 demonstrated that simple concatenation is not an effective fusion strategy for attribute-specific tasks. The introduction of the MoE model (Experiment 3) solves this limitation by dynamically learning the contribution weights per-image. The trainable router adapts to image content: images more discriminative by color induce high α; images more discriminative by texture induce high β. This behavior makes the MoE superior to concatenation in 7 of 8 configurations, and superior to or tied with single-layer extraction in 6 of 8 configurations.

Figure 5 shows this routing behavior for iBOT: the color-optimized model yields median $\alpha \approx 0.55$ (early layer), while the texture-optimized model yields $\alpha \approx 0.44$, confirming that the router allocates capacity according to the dominant attribute without explicit supervision.

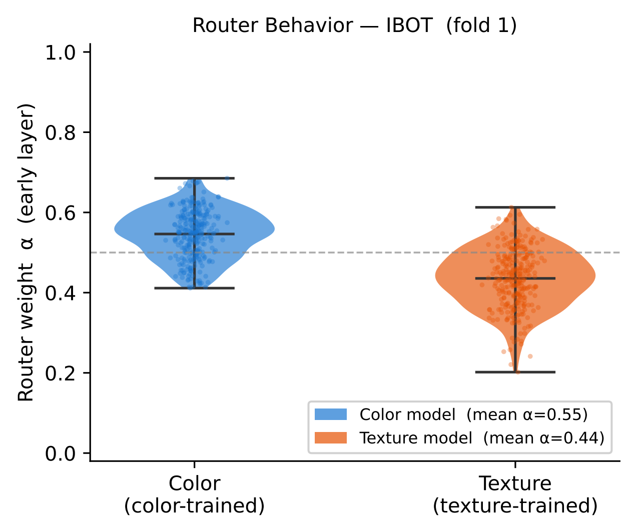

Fig. 5. Distribution of early-layer router weights α for color-trained and texture-trained MoE models (iBOT, fold 1). The color model assigns higher weight to the early layer (median $\alpha \approx 0.55$), while the texture model shifts toward the deep layer (median $\alpha \approx 0.44$). Distributions obtained with logit-space L2 regularization to prevent routing collapse.

The exception for iBOT texture (-2.40 pp over Exp. 1) is explained by the fact that Block 9 of iBOT already provides exceptionally rich texture representations (97.20%); the router cannot improve what is already near the dataset ceiling. For

VGG-16 on color, the dimensional asymmetry between the small early layer (64d) and the deep layer (256d) may limit the capacity of projector h_E.

In practical terms, the MoE offers a unified representation for visual-attribute retrieval systems, eliminating the need to manually select the most suitable layer for each query type.

The CBIR results provide additional evidence that the proposed adaptive fusion improves not only classification performance but also the structure of the embedding space. Across most backbones, increases in classification accuracy are accompanied by consistent gains in mAP@10, R@1, and R@5, indicating that semantically related images become more tightly clustered. Interestingly, the retrieval improvements are often larger than the corresponding classification gains, suggesting that the MoE enhances neighborhood organization beyond what classification accuracy alone captures. This effect is particularly visible for iBOT on the color dataset, where the MoE improves mAP@10 from 94.93% to 97.96% while simultaneously correcting the severe degradation introduced by simple concatenation. Conversely, in cases where a single specialized layer already achieves near-ceiling retrieval performance, such as iBOT Block 9 for texture, the MoE provides little additional benefit, indicating that the underlying representation is already highly discriminative.

VII. Conclusions

This work evaluated layer-wise feature extraction across five deep learning architectures for color and texture classification and proposed an MoE-based adaptive fusion model. The layer-wise sensitivity pattern observed, i.e., early layers for color and deep layers for texture, generalizes from CNNs to Vision Transformers and SSMs, and simple concatenation is not an effective fusion strategy. The MoE outperforms both baselines in 5 of 8 configurations and ties the single-layer baseline in one additional case, achieving 98.13% (VMamba, texture) and 98.13% (iBOT, color), and provides a unified representation that adapts to image content without manual layer selection. Future works envisioned shall focus on exploring more than two experts, cross-layer attention, and large-scale retrieval.

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